

Simple Linear Regression (Supervised Machine learning algorithms)

Regression $\rightarrow$ To establish a relationship b/w the two variables / more than two variables.

Supervised ML

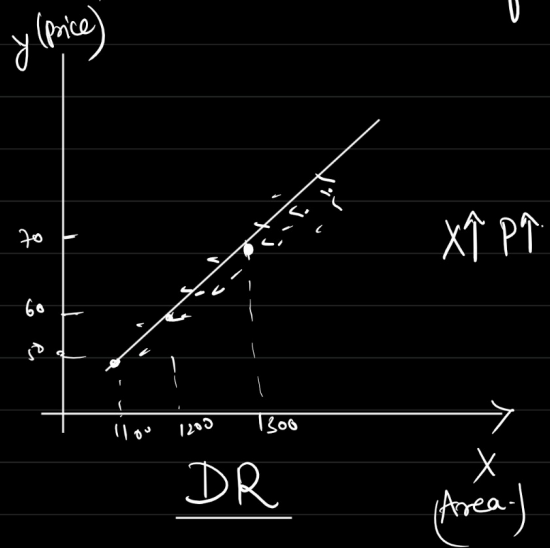
$\rightarrow$ Regression (y as continuous) Classification (discrete)

Linear $\rightarrow$ It establish a linear relationship.

* SLR attempts to determine the strength and characteristics of the relationship between one independent variable (X) and another dependent variable (Y)

* Predict the price of a house based on Area of house

Area of house (X)	Price of house (Y)
1100	50
1200	60
1300	70

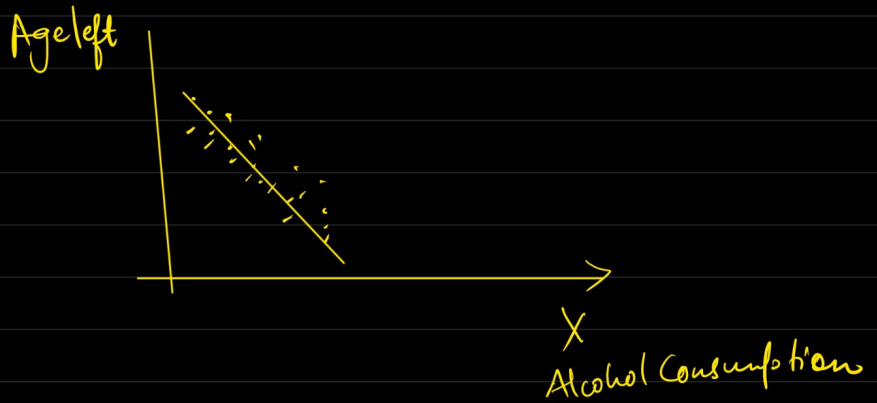


Ex: Selling Price of car wrt age of car

X (age of car)	Y (SP)
10	3.1
8	4.5
—	—



Age left Vs Alcohol Consumption



Simple

↓
Only one X and y

Linear

↓
Captures the linear relationship

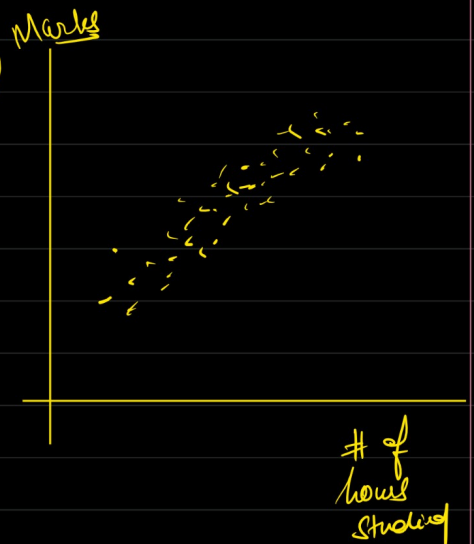
Regression

↓
Understand the relationship

* if multiple IV (multiple X's) → Multiple linear regression

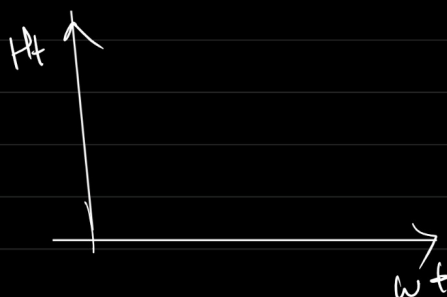
Example

No of hours studied	Marks obtained
8	72
9	85.5
—	—
—	—
—	—
—	—



Example

Weight	Height
65	168cm
70	172cm
—	—
—	—



* To predict price of house based on number of rooms.

y = price of house

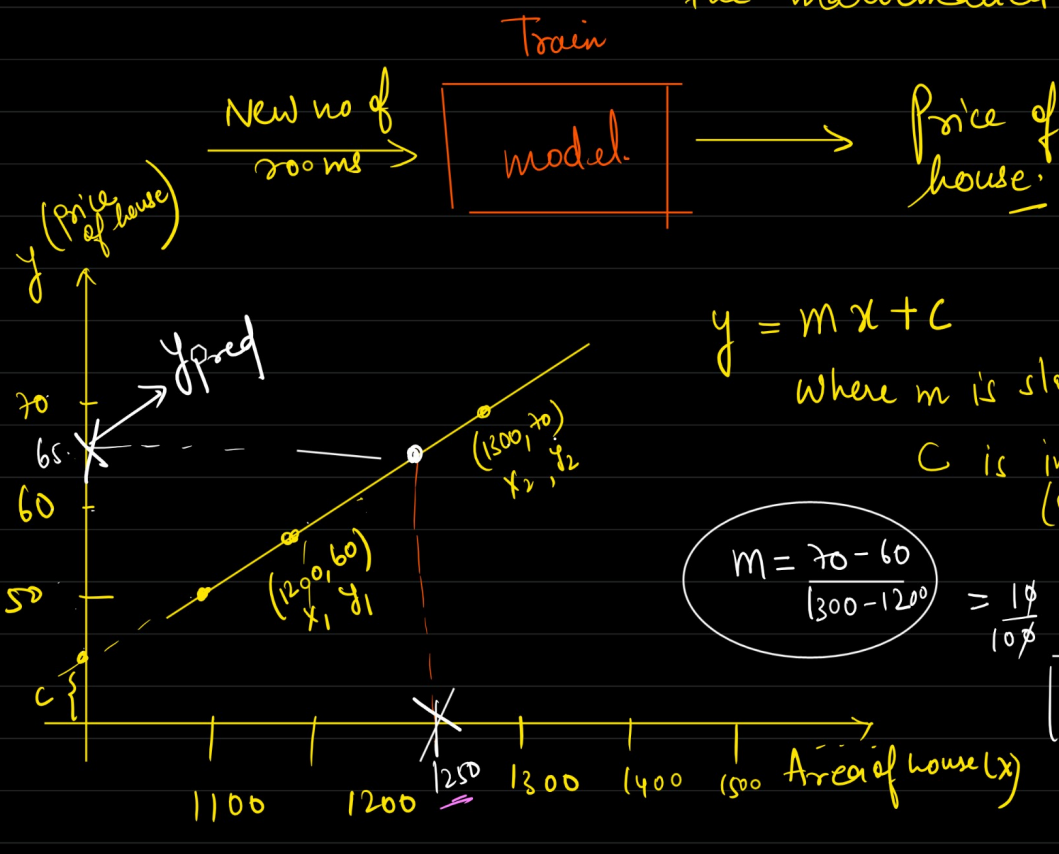
x = No of rooms.

# of rooms (x)	Price of house (y)
1	50
1.5	60
2	65
3	75
4	?? <u>85</u>

⇒ ML - To understand the mathematical relationship

This was 4 dp's that's why you understood the trend!

If many dp's you need to train the model to understand the mathematical relationship.



$$y = mx + c$$

$$y_{\text{pred}} = 0.1x + 10$$

↑
1250

$y_{\text{estimator}}$: -
You have estimated the price of a house

$$\hat{y} = mx + c$$

based on m , c , and Area of house

$$\hat{y} = mx + c$$

$$\hat{y} = \beta_0 + \beta_1 x \quad \left(\begin{matrix} \beta_1 = m \\ \beta_0 = c \end{matrix} \right) \rightarrow \text{Coefficients}$$

$$h_\theta(x) = \theta_0 + \theta_1 x$$

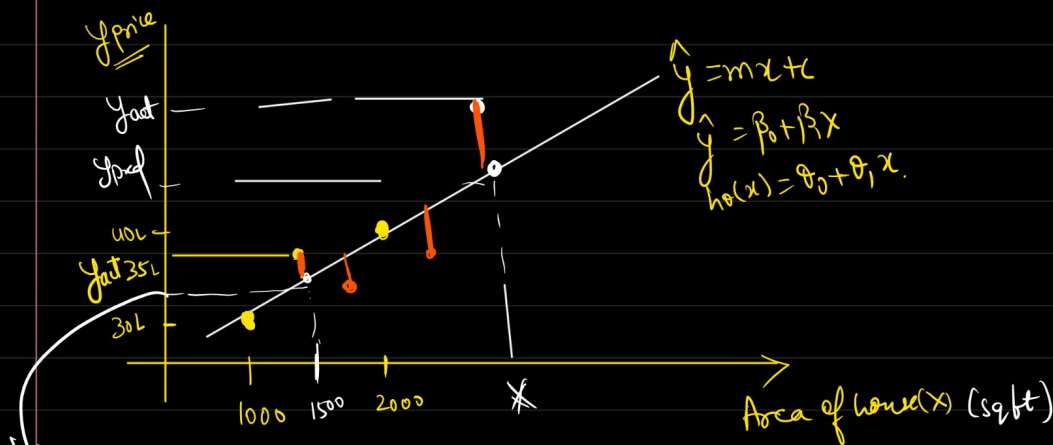
many books you will find this
hypothesis representation

because we consider a SLR line to be a hypothesis and it passes from these points.

$X \rightarrow IV$

$y \rightarrow DV$

$\left. \begin{matrix} m, c \\ \beta_0, \beta_1 \\ \theta_0, \theta_1 \end{matrix} \right\} \rightarrow \text{Coefficients}$

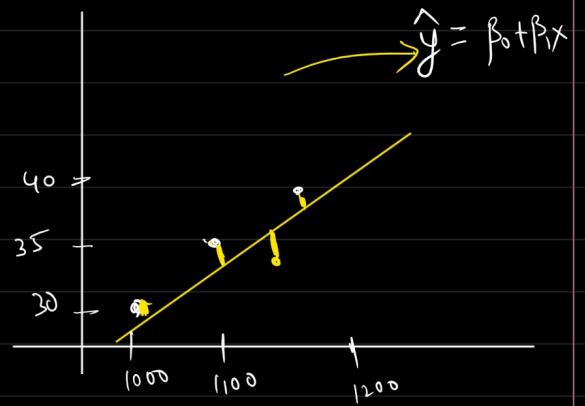


$y_{pred} = 33L$

$y_{actual} \text{ for } 1500 \text{ sq ft} = 35L$
 $y_{predicted} \text{ for } 1500 \text{ sq} = 33L$

$$\underline{\text{Error} = y_{act} - y_{pred.}}$$

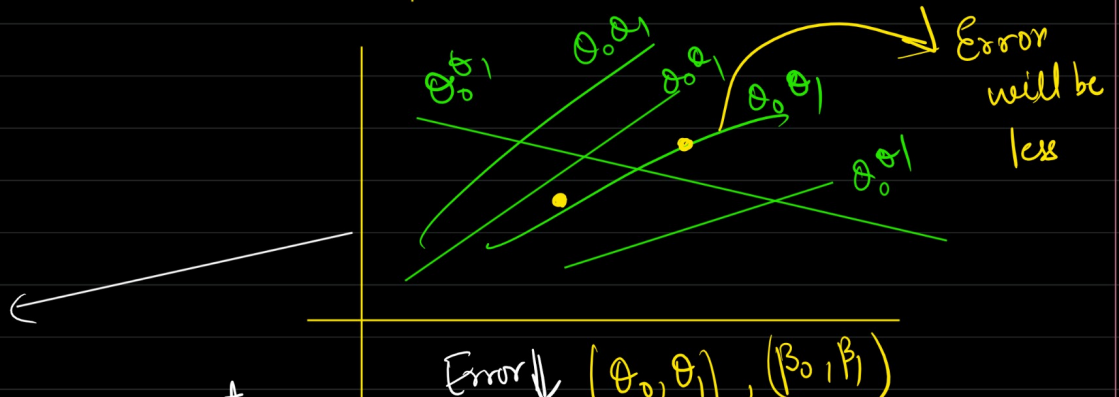
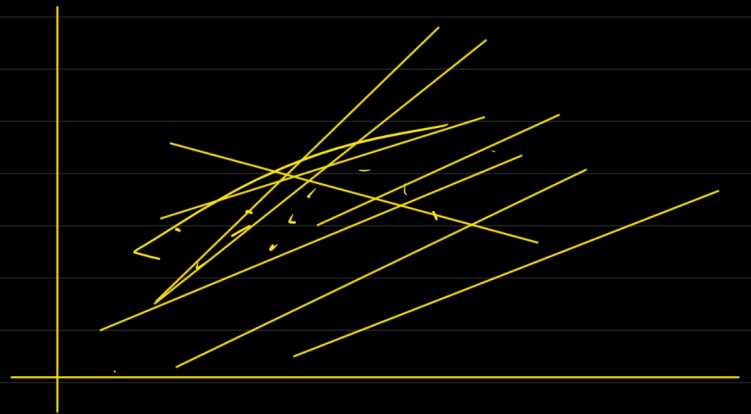
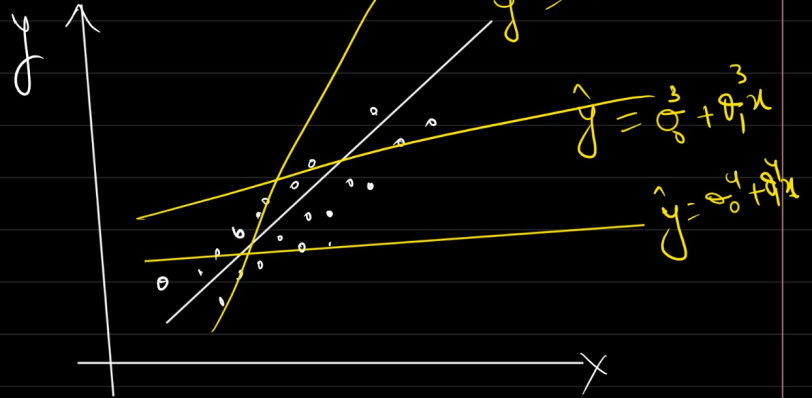
X (Area of house)	Y Price of house fact/ y_i	y_{pred}
→ 1000	30L	28L
→ 1100	35L	38L
→ 1200	40L	60L



$$\text{Error} = \text{factual} - y_{pred}$$

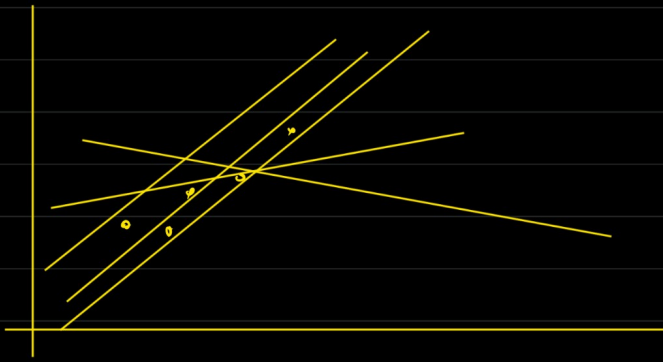
$$\text{Error} = y_i - \hat{y}_i$$

The predicted line should be made in such a way that overall sum of error should be minimum/least



for all lines, we want error = fact - y_{pred} to be least

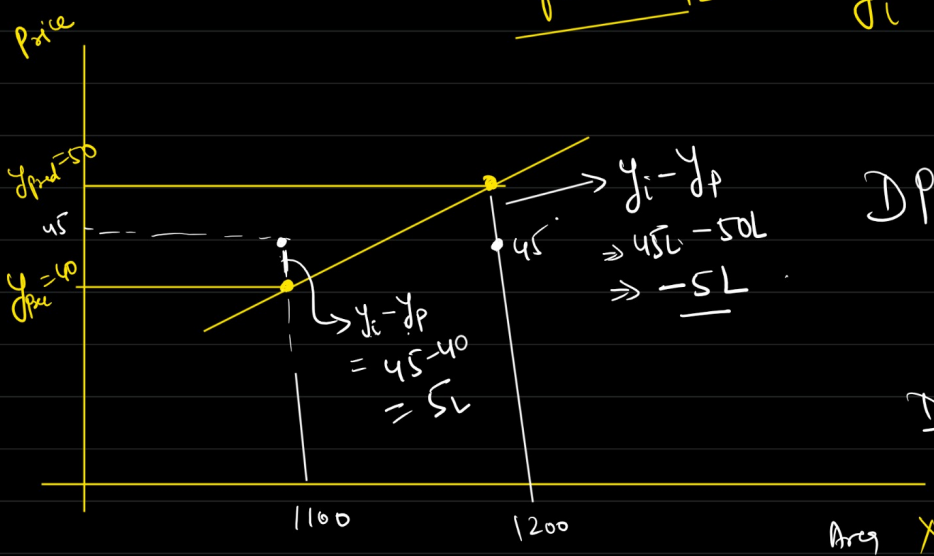
→ you want that coefficient where the error is least
→ optimal θ_0, θ_1



$(0, 0) \quad (0, 0) \quad (0, 0) \dots (0, 0)$
 $\Downarrow$
 Optimization process.

Error :- $y_{\text{act}} - y_{\text{pred}}$

for all dp_i : $y_i - y_{pi}$



$$\begin{aligned}
 DP_1 &= y_i - y_p \\
 &= 45 - 40 \\
 &= 5L
 \end{aligned}$$

$$\begin{aligned}
 DP_2 &= 45L - 50L \\
 &= -5L
 \end{aligned}$$

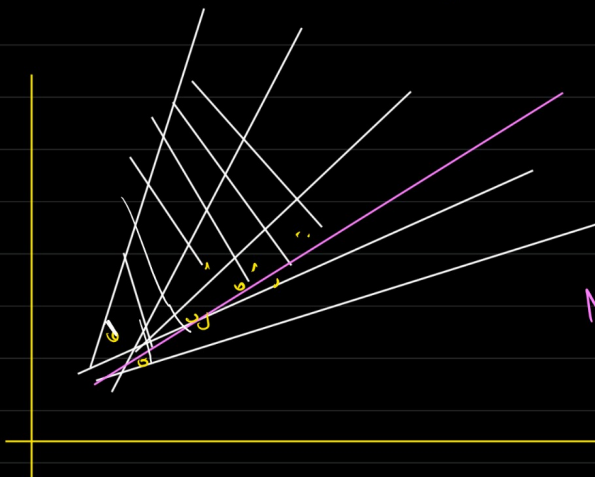
Error for all dp_i : $5L + (-5L) = 0$

$$\text{Squared Error} = (y_{\text{act}} - y_{\text{pred}})^2$$

$$\begin{aligned}
 &= (5)^2 + (-5)^2 \\
 &= 50
 \end{aligned}$$

Now you want that line where squared error is least

$$\min. (y_{\text{act}} - y_{\text{pred}})^2$$



$$\text{Total Error} = (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + (y_3 - \hat{y}_3)^2 + \dots + (y_n - \hat{y}_n)^2$$

$$\min \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\min \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\Rightarrow \min_{i=1}^n (y_i - \hat{y}_i)^2$$

$$h_0(x) = \theta_0 + \theta_1 x$$

$$\hat{y} = mx + c$$

$$\hat{y} = \beta_0 + \beta_1 x$$

$$\min \sum_{i=1}^n (y_i - h_0(x))^2$$

where n is no. of data point

least square method
or
Ordinary least square method

$$y_i = y_{\text{actual}}$$

$$h_0(x) = y_{\text{pred}}$$

$$\min \sum (y_i - h_0(x))^2$$

$$\min \sum_{i=1}^n (y_i - \theta_0 - \theta_1 x_i)^2$$

* we want to have that combination of m, c where the overall error is least and the line will be best fit line

$$\min_{i=1}^n (y_i - \theta_0 - \theta_1 x)^2$$

Technique to find best fit line
SLR = OLS

$$\begin{Bmatrix} \theta_0^1 & \theta_1^1 \\ \theta_0^2 & \theta_1^2 \\ \theta_0^3 & \theta_1^3 \end{Bmatrix} \rightarrow \text{Optimal } (\theta_0, \theta_1)$$

$$\begin{Bmatrix} 1 \\ \vdots \\ \theta_0 \\ \vdots \\ \theta_1 \end{Bmatrix}$$

$$\text{SSE} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$$

Sum of Squared Error

$$\left(\frac{1+2+3}{3} \right) \rightarrow \text{mean}$$

↓
Mean Sum of Squared Error
(MSE)

for best line
MSE should be least

MSE → Cost function
(penalty)

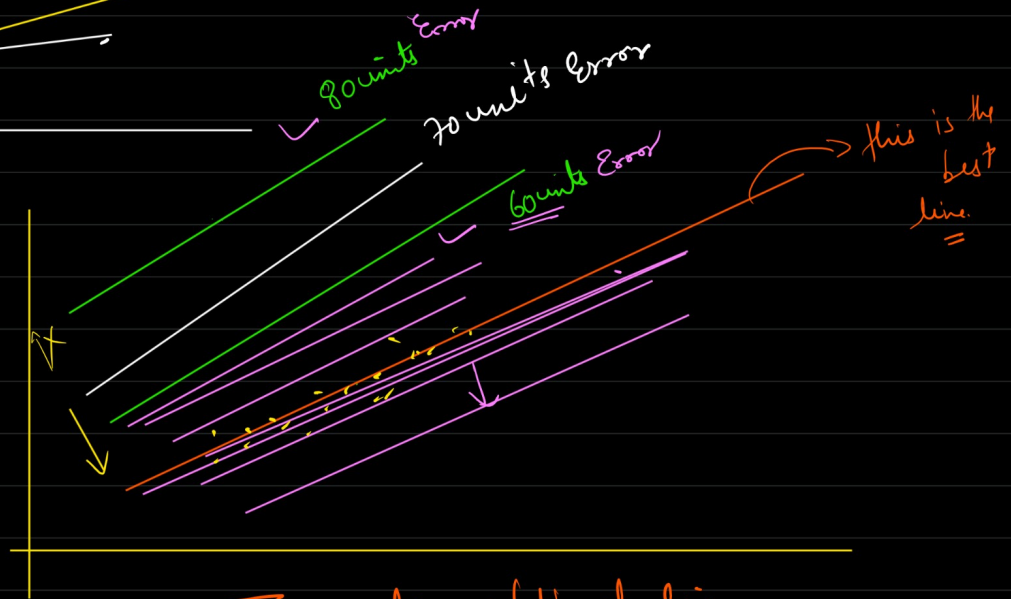
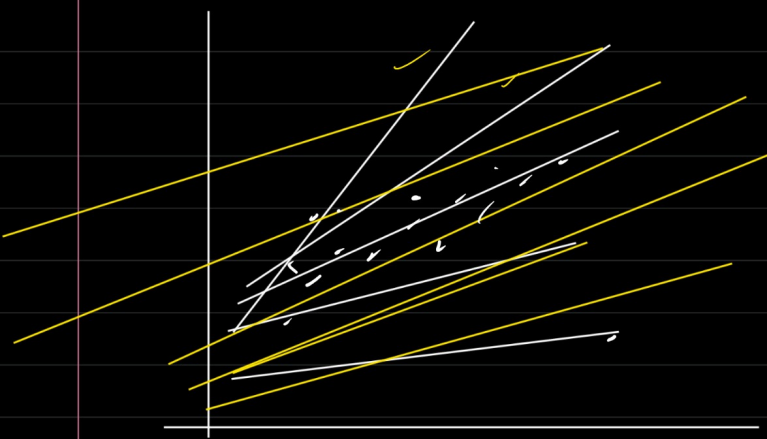
* MSE is used as CF to get optimal θ_0, θ_1 (optimal solution)

Why CF is Penalty?

Where SSE will be lowest?
→ 2nd line

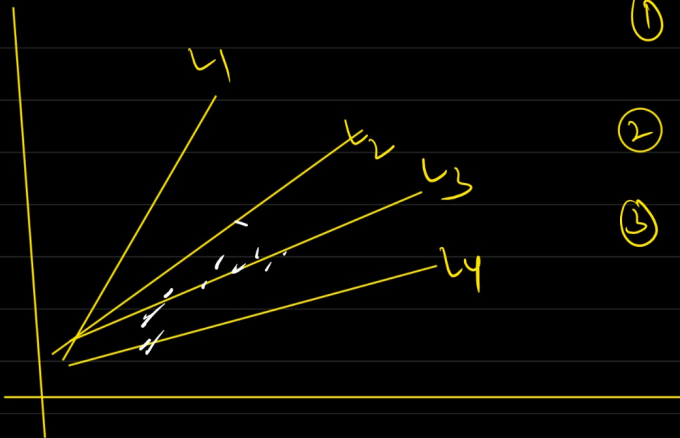


High MSE means High Penalty
↓
if you do right thing (in a game) you will not have penalty. Similarly if the best fit line passes close to all d.p's, less penalty will be there.



To get optimal line

- ① Chose right combination of line.
- ② Find Mean Squared error line
- ③ Find best fit line.



MSE is least
 $\Downarrow$
 best fit line.
 Optimal coefficients =

